Practice workbook · Explain sewage treatment, pit latrine design and decomposer roles

Stages of a sewage works, aerobic and anaerobic microbes, biogas and sludge, decomposers recycling nutrients, and how pit latrines are sited, built and covered.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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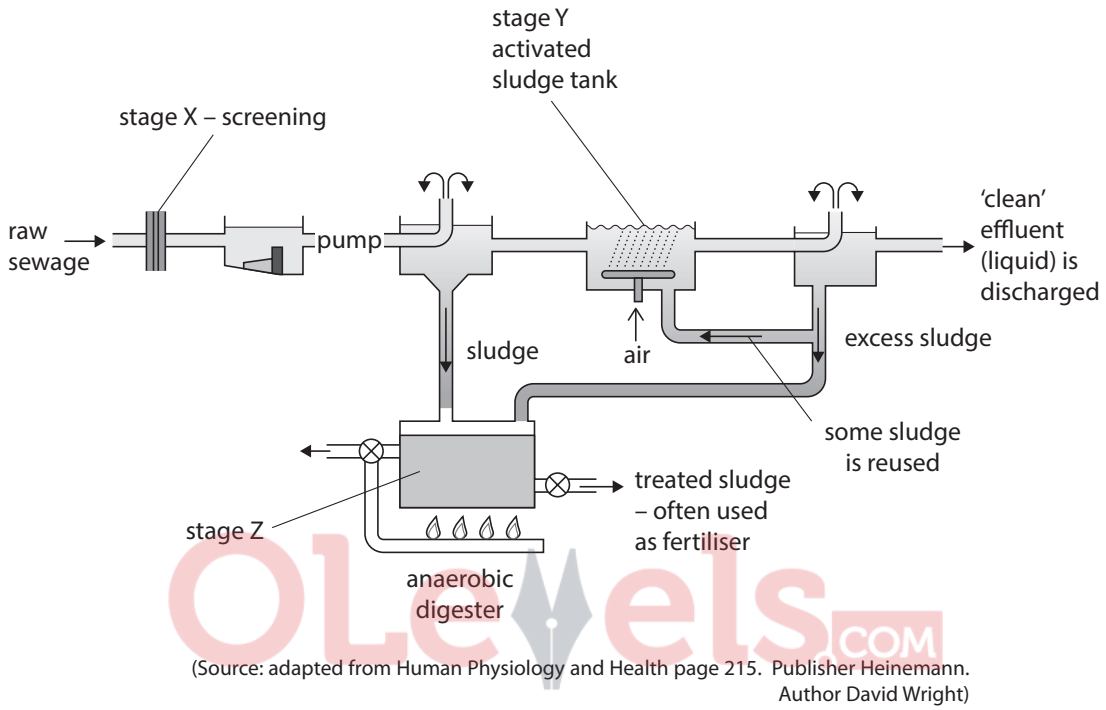
Type ID: HBI-SANITATION-SEWAGE-AND-LATRINES

1. Question 8 · part a.ii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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1. Question 8 · part a.ii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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1. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

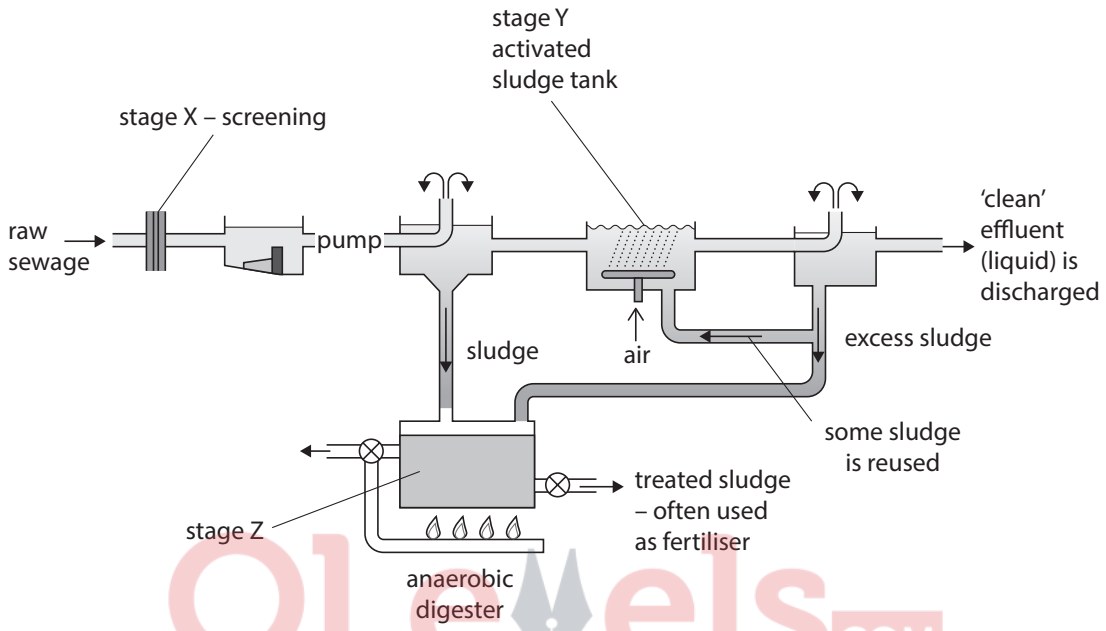
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2. Question 8 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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P 7 4 6 2 6 R A 0 2 0 2 4

2. Question 8 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

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(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

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(Total for Question 8 = 7 marks)

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2. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

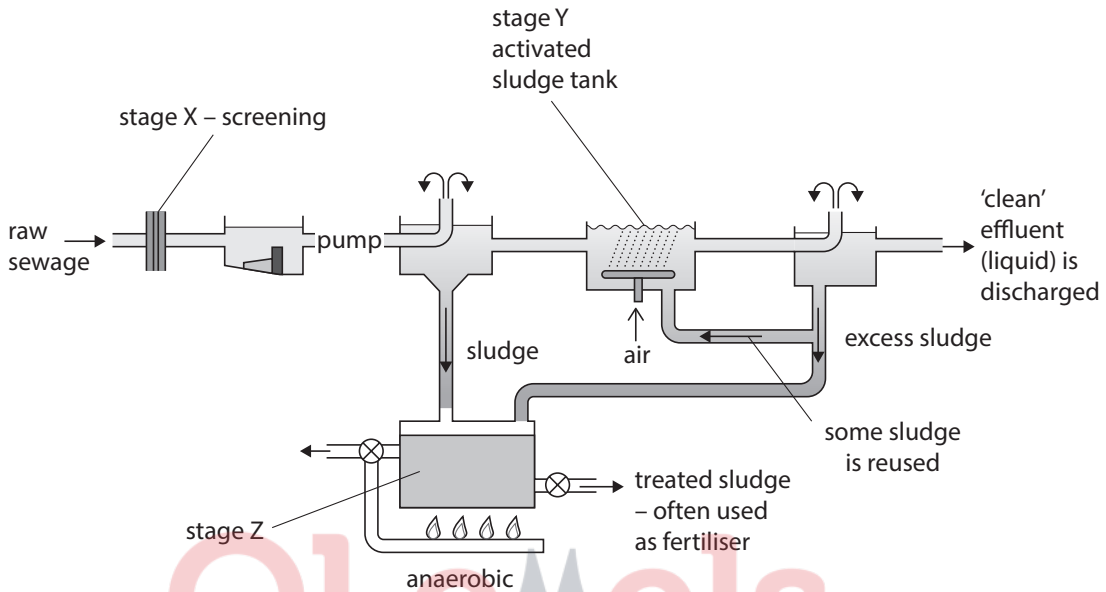
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3. Question 8 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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3. Question 8 · part a.iii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.iii only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

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(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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3. Question 8 · part a.iii · 1 mark · Working and answer

Working and answer

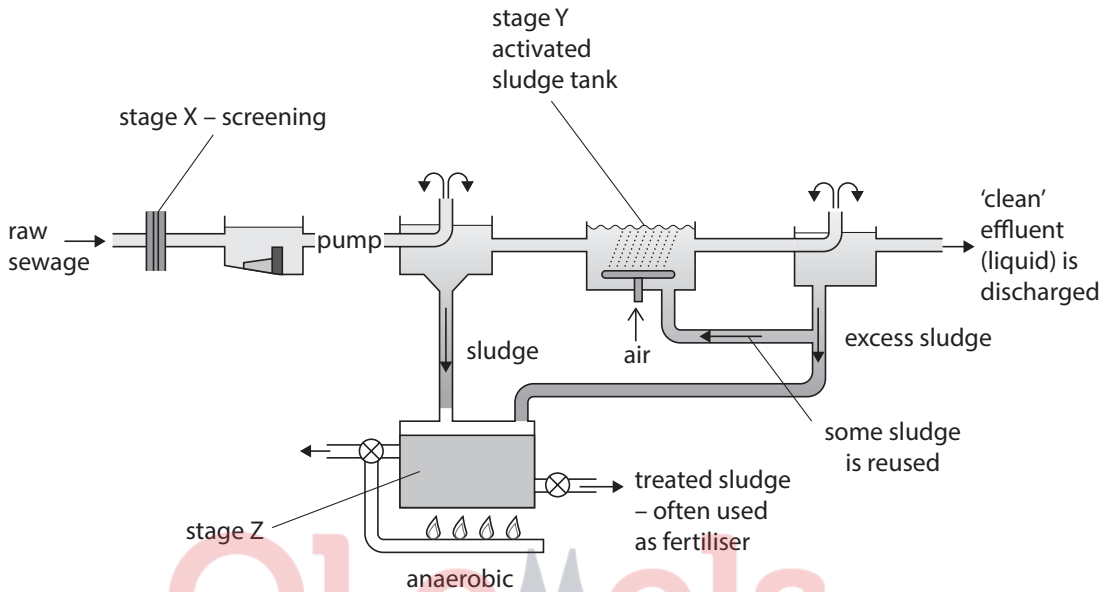
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4. Question 8 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

8 (a) The diagram represents a modern sewage works.



(Source: adapted from Human Physiology and Health page 215. Publisher Heinemann. Author David Wright)

(i) Describe the process that happens at stage X.

(2)

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P 7 4 6 2 6 R A 0 2 0 2 4

4. Question 8 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

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(ii) Explain why air is added at stage Y.

(2)

(iii) Biogas is produced at stage Z.

Name the main gas found in biogas.

(1)

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(b) Explain why treated sludge from a sewage works can be used as a fertiliser.

(2)

(Total for Question 8 = 7 marks)

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4. Question 8 · part b · 2 marks · Working and answer

Working and answer

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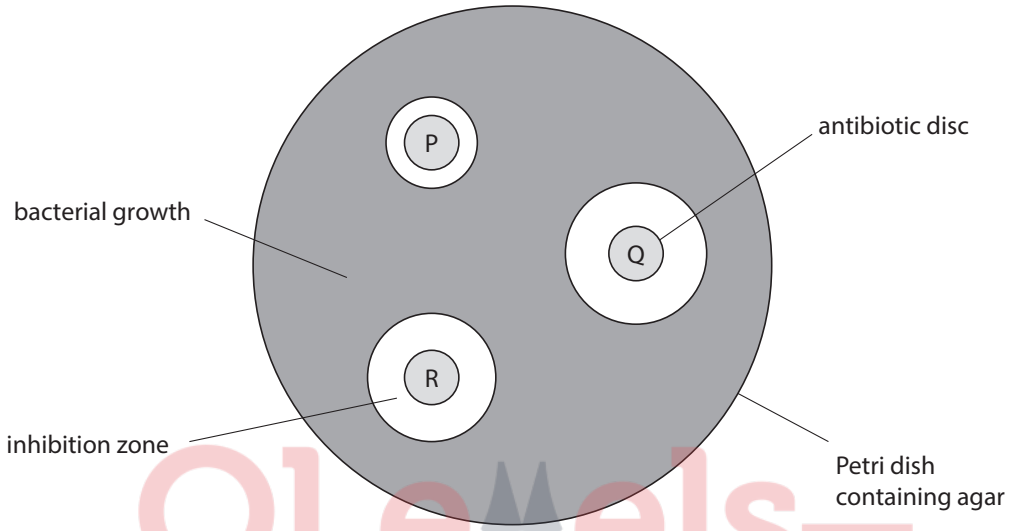
5. Question 5 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

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5. Question 5 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

5. Question 5 · part b · 2 marks · Working and answer

Working and answer

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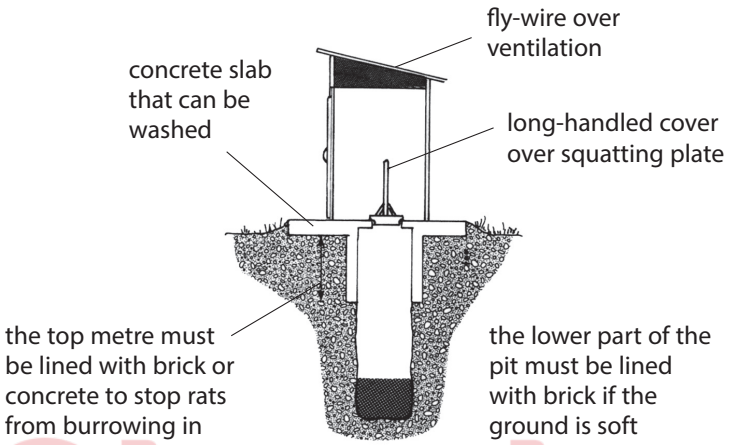
6. Question 5 · part a.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.i only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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6. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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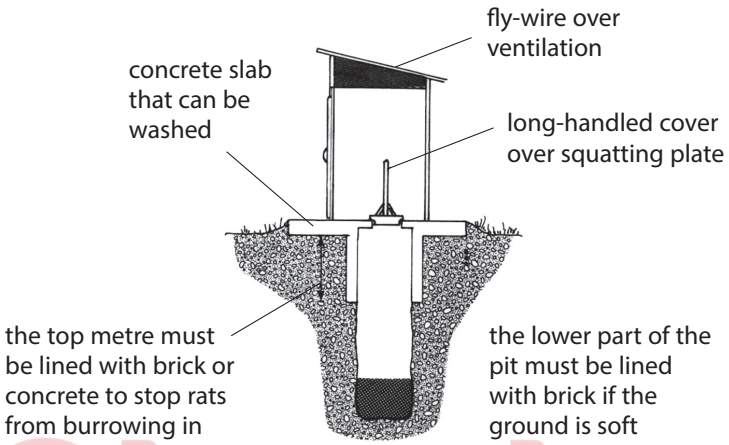
7. Question 5 · part a.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.ii only.

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7. Question 5 · part a.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part a.ii only.

(iii) Explain why, when a pit latrine is not in use, the latrine should have a cover and why the cover should have a long handle.

(5)

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7. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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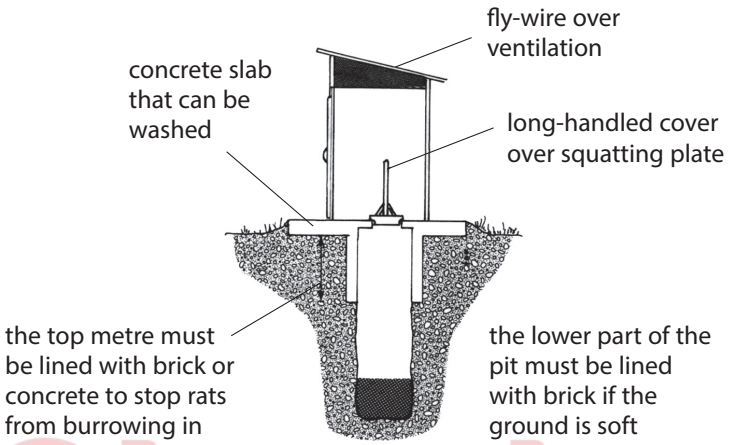
8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

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8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

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(5)

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8. Question 5 · part a.iii · 5 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(b) The concrete slab shown in the diagram is washed regularly with an antibacterial solution.

A scientist wants to investigate two antibacterial solutions to see which solution is more effective.

Design an investigation the scientist could use to find out which of the two antibacterial solutions is more effective.

Include experimental details in your answer and write in full sentences.

(7)

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(Total for Question 5 = 15 marks)

8. Question 5 · part a.iii · 5 marks · Working and answer

Working and answer

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